

Audrey Houghton (she/her)

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TECHNICAL SKILLS

Software

Proficient with Python, UNIX/Linux, GitHub, LaTeX, EMACS, VSCode, Singularity, Docker, and virtual environments. Familiar with Matlab, R, IDL, JavaScript, Node.js, npm, Go, Ansible, AWX, RedHat, ReadTheDocs, and Google Apps Script (via clasp). Knowledgeable in CI/CD pipelines (e.g., CircleCI, GitHub Actions) and deep learning AI (nnU-Net). Applied skills in DevOps, data engineering, and software engineering roles.

Data

Proficient in high-performance computing and data systems, including AWS (S3), Ceph, SLURM, Kubernetes, Globus, Grafana, and Prometheus. Experienced in ETL-based data warehousing, batch processing, SQL querying, and testing and validation of data pipelines. Skilled in statistical analysis, benchmarking, causal inference, predictive modeling, GIS-based spatial analysis, and electronic health record (EHR) data integration. Experienced in automation workflows including Salesforce API integration, Microsoft Teams automation, and compliant web scraping practices.

Product Management

Proficient in cross-functional and customer communication, Kanban boards, Jira, and documenting SOPs, methodologies, terminology, and codebases. Experienced with MVP development life cycles and familiar with Agile and Scrum frameworks. Applied business process automation to improve sales and reporting workflows.

EXPERIENCE

University of Minnesota – Department of Psychiatry & Behavioral Sciences

Dec 2025 - Present

Research Software Engineer III

Minneapolis, MN (Remote)

- Develop and maintain statistical and computational analysis pipelines for GIS-linked electronic health record data
- Implement causal inference, predictive modeling, and machine learning workflows in Python, R, and JavaScript
- Write and interpret results for publications and grant mechanisms, serving as a data and methods resource for an interdisciplinary research team

CoreWeave, Inc.

Apr 2024 - Sept 2024

Fleet Operations Engineer

Remote

- Provisioned and maintained thousands of NVIDIA GPU nodes in Kubernetes clusters using Ansible, AWX, Grafana, and Go, achieving onboarding timelines of 1–2 weeks with 6+ months average production uptime
- Diagnosed and resolved hardware/software issues across the node lifecycle, saving approximately \$250,000 per node repaired
- Partnered with 10+ teams—including networking, hardware, operations, and support—to streamline troubleshooting, accelerate incident resolution, and document SOPs and post-mortems, improving efficiency and reliability
- Identified points in the node lifecycle where nodes were lost, implementing solutions that saved the company millions in revenue

Masonic Institute of the Developing Brain

Jan 2021 - Apr 2024

Research Software Engineer II → Research Software Engineer III

Minneapolis, MN (Remote)

- Developed Python wrappers leveraging AWS S3 storage and SLURM optimization to accelerate image processing up to 12x
- Engineered a containerized infant MRI segmentation application with deep learning models, achieving 600x speed-ups and +53% accuracy
- Built ETL-based pipelines to transfer and process petabytes of imaging data 10x faster than prior industry benchmarks, incorporating statistical error analysis and validation to ensure reliability and reproducibility
- Authored comprehensive documentation and curriculum, setting standards for reproducible neuroimaging data processing

Calyxt

Oct 2019 - Aug 2020

Laboratory Assistant & Data Specialist

Roseville, MN

- Developed Python scripts for comprehensive data tracking and analysis methods

Huntington Learning Center

Aug 2019 - Oct 2019

Mathematics Instructor

Plymouth, MN

Tutor Doctor

Jan 2019 - Jun 2019

Mathematics Instructor

Loveland, Colorado

Waste Not, LLC.

Est. 2018

Data Manager and CEO

Amazon

Jul 2018 - Aug 2018

Data Entry Associate

Seattle, WA

Imagine Prep at Surprise

Mathematics Instructor

Jul 2017 - Jul 2018

Surprise, AZ

The MINVERA Project

Research Assistant

Jan 2015 - May 2017

Missoula, MT

- Developed a data reduction pipeline for time-domain imaging data written in Python

Rutgers University Astrophysics Department

Research Assistant

May 2016 - Aug 2016

New Brunswick, NJ

- Refactored IDL scripts into Python
- Generated models for galactic simulations through parameter optimization

PRODUCTS

Sales Summary Project

Automation Pipeline for Lead Generation & Sales Tracking

2025

[Proprietary]

- Designed a proof-of-concept pipeline integrating Salesforce logging, CSV rotation, and Microsoft Teams alerts
- Implemented compliant web scraping (robots.txt, throttling, retries) to demonstrate reliable, ethical data collection for potential enterprise use

JavaScript Excel Sheet App

Google Apps Script Utility for Dynamic Links

2025

[Personal]

- Built a Google Sheets App Script to dynamically generate links, clean outdated entries, and trigger updates in real time
- Reduced personal job application tracking effort by automating link formatting and updates; version-controlled with GitHub using clasp, Node.js, and npm

CoreWeave Internal Documentation

CoreWeave, Inc.

2024

[Proprietary]

- Initiated and built company-wide documentation from the ground up, replacing scattered one-off documents with an organized, centralized Notion knowledge base covering terminology, SOPs, and team workflows

SLURM Wrappers

High-Performance Computing S3 and Disk SLURM-Compliant Wrappers

2021 - 2024

github.com/DCAN-Labs/SLURM_wrappers

- Bash/Python wrappers adaptable to any SLURM-based data pipeline

ABCD Study Collection 3165 Data Processing

NIH Data Archive (NDA)

2021 - 2024

collection3165.readthedocs.io

- Community dataset enabling access to petabytes of processed/unprocessed MRI data under BIDS standards

DCAN Labs Informational Guide

Minnesota Supercomputing Institute Data Engineering Documentation

2022 - 2024

dcan-labs-informational-guide.readthedocs.io

- Comprehensive, continually-supported documentation of usage of the Minnesota Supercomputing Institute to process, analyze, and transfer MRI data

CABINET

Container-Linking Wrapper

2022 - 2023

github.com/DCAN-Labs/CABINET

- Python-based universal container-linking wrapper employing FAIR principles

BIBSNet

Infant MRI Brain Segmentation Application

2022 - 2023

github.com/DCAN-Labs/BIBSnet

- Containerized infant MRI application; achieved 600x faster processing and +53% accuracy

Data Processing Workshop

Minnesota Supercomputing Institute (MSI) Data Processing Workshop

2022

tinyurl.com/MRIsyllabus

- An extensive all-levels MSI data processing workshop
- Used as a template for a University neuroscience course at the University of Minnesota

EDUCATION

University of Montana - Missoula, MT

Bachelors of Arts in Physics

Aug 2013 - May 2017

umt.edu/physics-astronomy

University of Minnesota

Data Management Series

Fall 2023

libguides.umn.edu/datamanagement/dmbootcamp

University of Minnesota

Project Management Certificate Coursework (In Progress)

2023

ccaps.umn.edu/project-management-certificate

Completed modules: Foundations, Execution & Monitoring, Leadership, Team Communication